

Factor Exposures and Hedge Fund Operational Risk: The Case of Amaranth

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Abstract

Hedge fund performance and risk measurement continues to present intriguing challenges to both academics and practitioners. Risk-return measures that are solely based on historical return series tend to provide limited information and the marginal new information revealed by another quantitative measure tends to be small, and approaches zero once three or more measures are considered. In this article we will examine the risk exposures and performance characteristics of Amaranth Advisors LLC. Amaranth Advisors LLC was created in 2000 as a multi-strategy hedge fund. Beginning operations with approximately \$600 million in capital, it sought to employ a diverse group of arbitrage trading strategies particularly featuring convertible bonds, mergers and utilities. In 2002, Amaranth added energy commodity trading to its slate of strategies with JP Morgan Chase clearing its commodity trades. On August 4, 2006 NYMEX examined Amaranth's positions and calculated that Amaranth held about 51% of the open interest in the September natural gas futures contract which would expire at the end of the month. NYMEX decided that this was too large and on August 8 NYMEX compliance officials notified Amaranth of their concerns. Amaranth complied with NYMEX's directions and subsequently reduced its September and October positions. However, at the same time Amaranth increased its positions in September and October ICE contracts such that their overall positions in natural gas rose. The events that followed in late August and September led to huge losses with Amaranth losing significant value.

INTRODUCTION

Hedge fund performance and risk measurement continues to present intriguing challenges to both academics and practitioners (Kazemi and Schneeweis [2003]). Gupta [2005] examines the interdependence of several measures for various hedge fund strategies and concludes that in most cases these measures do not reveal any unique information and portfolio selection based on many of these measures result in highly correlated portfolios. This implies that risk-return measures that are solely based on historical return series tend to provide limited information and the marginal new information revealed by an additional quantitative measure tends to be small, and approaches zero once three or more measures are considered.

Further, quantitative risk-return measures that are solely based on historical returns very often fail to provide the tools that investors need to protect themselves against serious losses. One of the main reasons for this failure is the uniqueness of hedge fund strategies and related operational issues in executing each strategy. Consider, for example, the recent problems at Amaranth Advisors, LLC. Even though the firm emphasized that its fund was multi-strategy, most of the recent losses were driven by adverse natural gas trades. Prior to this debacle, most investors who viewed reports from this firm saw no reason to worry about its performance although some privy to their portfolio positions expressed concern. The compound annual return for the period September 2000 – November 2005 according to media reports (Morgensen and Anderson [2006]) was 14.72% net of all costs. Clearly, knowledge of the fund's returns did not enable investors to assess the fund's unique risks correctly. Note, however, that Amaranth's misfortunes

were solely a result of poor risk management. Note, also that even though the firm lost over \$6 billion in a matter of days, the losses had minimal impact in the industry as a whole.

Consider another example, the case of Bayou Funds. This case is among the most highly publicized in recent years. Exhibit 1 lists the summary statistics of Bayou's reported returns as well as those of comparable CISDM, CSFB and HFR indices. While the indices reported annualized returns in the 4-7% range over the period Jan 2000 – Dec 2004, Bayou reported a return of 13.14% over that same period. Moreover the volatility of Bayou's returns was significantly lower than those of the three indices. The primary question that arises from this discussion is how relevant are the return figures? The return figures are extremely important as a first check in the quantitative due diligence process. However, the returns do not reveal the entire risk profile. A rigorous qualitative due diligence process is required to supplement the returns-based analysis. Interestingly, according to media reports, since diligence conducted by certain firms revealed problems at Bayou and some investors chose not to invest in the fund.

<< Exhibit 1 Here >>

In this article we will examine the risk exposures and performance characteristics of Amaranth Advisors LLC. We will describe the specific characteristics of the failure in the next section.

Below we briefly review some previous work in the area of risk measurement and hedge fund defaults. In the next section we discuss our database and methodology followed by a section reporting our results. Finally, we will offer some concluding thoughts on the implications of our analyses and issues for future research.

Recent research has focused on quantifying and valuing the disclosures of various hedge fund default risks. These risks may be in various forms such as financial, operational and straight fraud risks. Corentin, Daul and Giraud [2006] examine the statistical properties of 109 defaults over the period 1994-2005. They find that a high percentage of defaults (over 50%) were due to fraud compared to defaults due to operational risk (13%) or defaults due to financial risk (33%). Another interesting finding in the study is that a high percentage of funds (almost 50%) that defaulted had less than \$100 million under management. Further, 56% of defaults were in funds that traded simple instruments compared to 29% of defaults in funds that traded complicated or opaque instruments. Brown et.al [2007] use the recent controversial and unsuccessful SEC attempt to increase hedge fund disclosure to examine the potential benefits of registration to investors. They note that any consideration of disclosure requirements should take into account the endogenous production of information within the industry and the marginal benefit of required disclosure on different clienteles. Other research in the area of performance and risk measurement have focused on specific failures such as Amaranth (Martin [2007], Chincarini [2006]) as well as liquidity issues, Derman[2006]). For overviews of this line research, the reader is directed to Till and Gunzberg [2005] or Gehin [2006].

THE FAILURE OF AMARANTH LLC

The hearings on natural gas speculation by the permanent subcommittee on investigations of the Senate Committee on Homeland Security and Governmental Affairs clearly demonstrate that the Amaranth debacle could have been easily avoided had ICE like NYMEX had the ability to limit Amaranth positions.

Amaranth Advisors LLC was created in 2000 as a multi-strategy hedge fund with approximately \$600 million in capital. It sought to employ a diverse group of arbitrage trading strategies particularly featuring convertible bonds, mergers and utilities. In 2002, Amaranth added energy commodity trading to its slate of strategies with JP Morgan Chase clearing Amaranth's commodity trades. Amaranth returns are plotted against the CISDM Equal Weighted Hedge Fund Index and CISDM Convertible Arbitrage Index in Exhibit 2 and growth of \$1 in each of these indexes is plotted against growth of \$1 in Amaranth LLC in Exhibit 3.

<<< Insert Exhibits 2 and 3 Here >>>

Exhibits 2 and 3 reveal several points. First, the returns tracked both of these indices until May 2005. This is consistent with the finding that Amaranth did not assume significant energy positions until May 2005. Second the growth of \$1 in the CISDM Equal Weighted Hedge Fund Index also tracked that of Amaranth LLC until May 2005. On August 4, 2006 NYMEX examined Amaranth's positions and calculated that Amaranth held about 51% of the open interest in the September natural gas futures contract which would

expire at the end of the month. NYMEX determined that this was too large and on August 8 NYMEX compliance officials notified Amaranth of their concerns. During a follow up notification on August 9 NYMEX further notified Amaranth that they should not reduce September positions simply by shifting those positions to the October contract. Amaranth complied with NYMEX's directions and subsequently reduced its September and October positions. However, at the same time Amaranth increased its positions in September and October in ICE contracts and as a results increased their overall positions in natural gas. The events that followed in late August and September led to huge losses with Amaranth losing significant value.

We had noted earlier that knowledge of the Amaranth's returns did not enable investors to assess the fund's unique risks correctly. In this article we will show that careful quantitative due diligence would have revealed some interesting facts about the Amaranth portfolio. First, it would have confirmed as do Exhibits 2 and 3 the structural breaks in its return patterns in mid 2005 as well as the stellar returns generated in late 2005 and early 2006 from a clear change in strategy. Second this information coupled with the fact that they were being directed by NYMEX to reduce their natural gas positions would have caused significant concern to an investor, were the investor privy to this information. Third it also raises the broader question whether fund management should be obligated immediately to reveal to investors such notice of action from market regulators. While hindsight is 20/20, some investors might have elected to redeem their investments had they been fully apprised of the situation. Another important issue that is being debated is whether ICE should have the same authority as NYMEX to ask

speculators to decrease their positions if the size of those positions became of concern to the exchange. Clearly the Amaranth debacle could have been easily avoided had ICE had the same authority as NYMEX to direct Amaranth to decrease its positions. It is also noteworthy that Shane Lee on being pressed by Senator Levin during his testimony revealed that they would not have engaged in such trade transfers had they not been allowed to do so.

DATA AND METHODOLOGY

To the best of our knowledge, this is among the first articles that examine actual Amaranth returns. Prior articles on Amaranth have examined fund of funds that had investments in Amaranth (Martin[2007]) while others have examined Amaranth's positions (Chincarini [2007] and Till[2007]). In July 2007, the permanent subcommittee on Investigations of the Senate Homeland Security and Governmental Affairs Committee issued a report entitled "Excessive Speculation in the Natural Gas Market" which examined the Amaranth failure in detail. We obtained monthly return data for Amaranth from the permanent subcommittee¹. Hedge Fund indices returns were obtained from CISDM as well as other respective websites.

We will first present summary statistics for Amaranth. Next we will present single factor regression results against the relevant indices. Finally we will perform a "monitoring test" as described below to detect any structural breaks in the data.

¹ We thank Dan Berkovitz, Counsel, Permanent Subcommittee on Investigations for kindly providing us monthly return information in Amaranth.

When an investor decides to allocate capital to one or more hedge fund strategies, the first step is to perform due diligence. During this process the investor can learn a great deal about the fund, portfolio manager, his/her trading strategy, the fund's back office support and its risk management techniques. Once the due diligence is over and the commitment is made, the investor will typically visit the manager every few months to ensure that the provisions of the investment agreement are still followed. Between regular visits to the office, the investor has to rely on reported fund's returns to determine if the fund is following the agreed guidelines. In the last section of this paper we apply the "monitoring test" of Zeileis et.al. (2005) to show how between due diligence visits a fund's return could be used to determine if the manager has deviated from his/her investment mandate.

The mathematical details of the monitoring test can be found in the original paper of Zeileis et.al. (2005). Briefly, suppose the rate of return on a fund's strategy can be expressed by a linear factor model of the following form

$$r_t = \beta_0 + \sum_{j=1}^F \beta_j f_{jt} + \varepsilon_t \quad (1)$$

Suppose we have confidence that the manager's return up to the last due diligence visit, $t = n$, was generated by the above model. Therefore, our objective is to use returns observed at time t , where $t = n+1, \dots, T$, with T being the time for next visit, to determine if there has been change in any of the factor exposures. The monitoring test proposed by Zeileis et.al. (2005) uses the following test statistic

$$Y(t) = \Lambda(t) \times (\hat{\beta}^{(t)} - \hat{\beta}^{(n)}),$$

where $\hat{\beta}^{(n)}$ is the estimated value of the vector of the coefficients of the linear factor model using data up to period n and $\Lambda(t)$ is an scaling factor (see equation (3) of Zeileis et.al. (2005)). It can be shown that the statistic $Y(t)$ follows a Brownian bridge process.

To decide if there has been a structural change in the linear factor model the value of $Y(t)$ is compared to $\pm b(t)$, where

$$b(\tau) = \sqrt{\tau(\tau-1) \left[\lambda^2 + \log\left(\frac{\tau}{\tau-1}\right) \right]},$$

$$\tau = \frac{t}{n}$$

If $Y(t)$, for $t = n+1, \dots, T$, crosses the two boundaries, then the null hypothesis of no change is rejected. The value of λ^2 is selected to reflect the level of confidence (e.g., at 95% confidence level $\lambda^2 = 7.78$). We apply the above monitoring case to Amaranth LLC.

EMPIRICAL RESULTS AND DISCUSSION

Our results are segmented into three parts. First, we will present summary statistics for the Amaranth. Next we will present single factor regression results against the relevant indices. Finally we will present the results of the “monitoring test” of Zeileis et.al. (2005) which illustrate how a fund’s return could be used to determine if the manager has deviated from his/her investment mandate.

Summary Statistics

The summary statistics for Amaranth LLC are presented in Exhibit 4. The top panel examines the period Jan 2003 – Sept 2006, the middle panel examines the period Jan 2003 – Aug 2006 and the bottom panel examines the period Jun 2005 – Aug 2006. For all three periods it is clear that the volatility of Amaranth is much higher than both the CISDM Equal Weighted Hedge Fund Index as well as the CISDM Convertible Arbitrage Index. However the bottom panel further establishes the return patterns of Amaranth were significantly different from the patterns of a multi-strategy hedge fund. Another interesting feature is the magnitude of the kurtosis numbers. It is very clear that the fat tails establish that the return patterns of an extremely volatile asset such as natural gas but certainly not of a multi-strategy hedge fund. Any investor with access to these numbers could have determined this from simple analyses.

Single Factor Regressions

We performed regression analyses using Amaranth returns. We examined the returns of fund of funds that had invested in Amaranth against Amaranth's returns. The sample of fund of funds that was used was the same sample that was used in Martin [2007]. Martin [2007] examines 80 funds that had invested in Amaranth. In this article we use 65 funds of the sample of 80 that had complete return information for the period Jan 2003 – Aug 2006. We also regressed Amaranth's returns against the three major indices: CISDM Equal Weighted Hedge Fund Index, the CSFB Hedge Fund Index and the HFR Fund

Weighted Index. If Amaranth was truly a multi-strategy fund then single-factor regressions should reflect exposures to these indices.

The single factor regression R-squares of the fund of funds against Amaranth returns in displayed in Exhibit 5. The R-squares range from 0.13 – 0.88. The correlation of the R-squares with the September 2006 returns was -0.59. As expected and as is clear from the chart, those with low R-squares had higher returns than those with higher R-squares.

Exhibit 6 displays the frequency of R-squares and the average returns of those funds in that group. Again it is clear that the higher the R-square the greater the loss. These results reveal some interesting points. First, even though fund of funds are supposed to be well diversified and not have extreme exposures it is clear that Amaranth represented significant investments in many fund of fund portfolios. Investors may not have invested in these funds were they aware of the high exposures to Amaranth.

The single-factor regression results against the three major indices are displayed in Exhibit 7. In all three cases, both the R-Square and adjusted R-Squares are extremely low. Further the beta coefficients are not significant as indicated by the t-statistics and p-values. One would expect some exposure to these three major indices if the fund was truly exposed to various strategies.

Monitoring Test

In this section we apply the monitoring test discussed in the previous part to the return stream of Amaranth Advisors LLC. As discussed previously, Amaranth failed due to the large size of its natural gas positions on ICE. Our goal is to determine if the monitoring

test can be used to detect significant changes in these funds strategies after the first 12 months of operations. Typically, an investor will perform this test after the due diligence process is completed. In our case we assume that any significant changes in these funds strategies have taken place after the first 12 months of operations.

Using monthly figures on Lehman High Yield Bond Index, Goldman Sachs Commodity Index, Fama and French factors [1992]⁷ (excess return on market portfolio, small minus big and high book value minus low book value) and momentum factor of Carhart [1997]³, we estimated factor exposures that appear in equation (1).

<< Exhibits 8 >>

Exhibit 8 displays the results of our tests. The graphs display dynamics of standardized factor exposures of Amaranth to the above factors and the 95% confidence bands. If one of factor exposures moves outside the band, then one can conclude that the manager's style has significantly changed.

We can see that there were significant changes in the strategy after the first 12 months. The Senate report and our prior analysis indicated that Amaranth increased in energy positions around May 2005, and as it can be seen, the test indicates that the fund's factor exposures violated the 95% boundaries.²

² The above monitoring test was applied by Gupta and Kazemi [2007] to a sample of live funds. The results show that the test gives "false negatives" in about 12% of cases. Therefore, the results of this test should be used rather carefully and as stated in the paper, in most cases on-site due diligence is the most effective way that an investor can verify if a hedge fund is adhering to its stated objectives.

CONCLUSIONS

One of the continuing challenges to hedge fund performance measurement is the efficacy of available measures. In this article we examined the risk exposures and performance characteristics of Amaranth Advisors LLC. We found that the return patterns of Amaranth were significantly different from the patterns of a multi-strategy hedge fund. We also established that any investor with access to these numbers could have determined this from simple analyses. Further we conducted regression analyses of fund of funds against Amaranth returns. These results revealed some interesting points. First, even though fund of funds are supposed to be well diversified and not have extreme exposures to any one fund it is clear that Amaranth represented significant investments in many fund of fund portfolios. Investors may not have invested in these funds were they aware of the high exposures to Amaranth.

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Exhibit 1: Summary Statistics of Bayou Funds

	Annualized Return	Standard Deviation	Skew	Kurtosis
BAYOU FUNDS	13.14%	4.58%	-1.01	2.67
CSFB-Tremont HFI Long-Short Equity	4.83%	9.01%	0.79	5.11
HFRI Equity Hedge Index	6.26%	8.75%	0.67	2.21
CISDM Equity Long/Short Index	5.73%	6.64%	0.42	0.80

Exhibit 2: Amaranth Returns

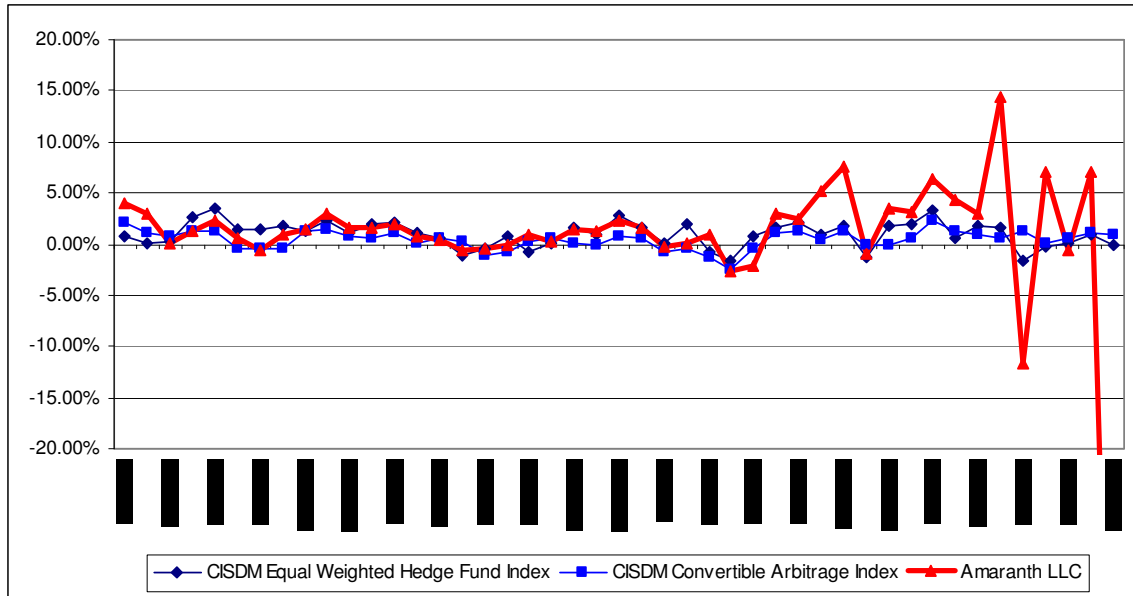


Exhibit 3: Growth of \$1 Invested in Amaranth

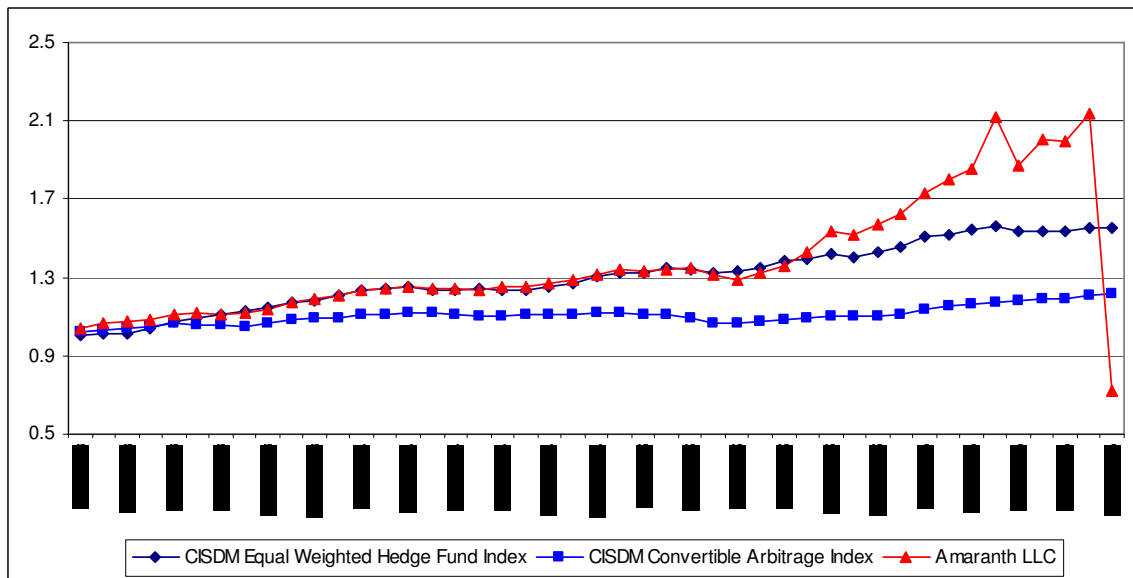


Exhibit 4: Amaranth Summary Statistics

	CISDM Equal Weighted Hedge Fund Index	CISDM Convertible Arbitrage Index	Amaranth Advisors LLC
Annualized Return	11.82%	5.31%	3.54%
Annualized Standard Deviation	4.33%	3.18%	37.16%
Skewness	-0.30	-0.72	-5.56
Kurtosis	-0.29	1.09	34.91
Minimum	-1.68%	-2.45%	-66.00%
Maximum	3.54%	2.34%	14.42%
	CISDM Equal Weighted Hedge Fund Index	CISDM Convertible Arbitrage Index	Amaranth Advisors LLC
Annualized Return	12.09%	5.18%	21.62%
Annualized Standard Deviation	4.34%	3.21%	12.60%
Skewness	-0.35	-0.69	-0.02
Kurtosis	-0.24	1.01	6.29
Minimum	-1.68%	-2.45%	-11.66%
Maximum	3.54%	2.34%	14.42%
	CISDM Equal Weighted Hedge Fund Index	CISDM Convertible Arbitrage Index	Amaranth Advisors LLC
Annualized Return	12.23%	10.23%	42.98%
Annualized Standard Deviation	4.67%	2.24%	19.41%
Skewness	-0.68	0.38	-1.06
Kurtosis	0.16	0.73	3.93
Minimum	-1.68%	-0.17%	-11.66%
Maximum	3.28%	2.34%	14.42%

Exhibit 5: Single-Factor Regression R-Squares

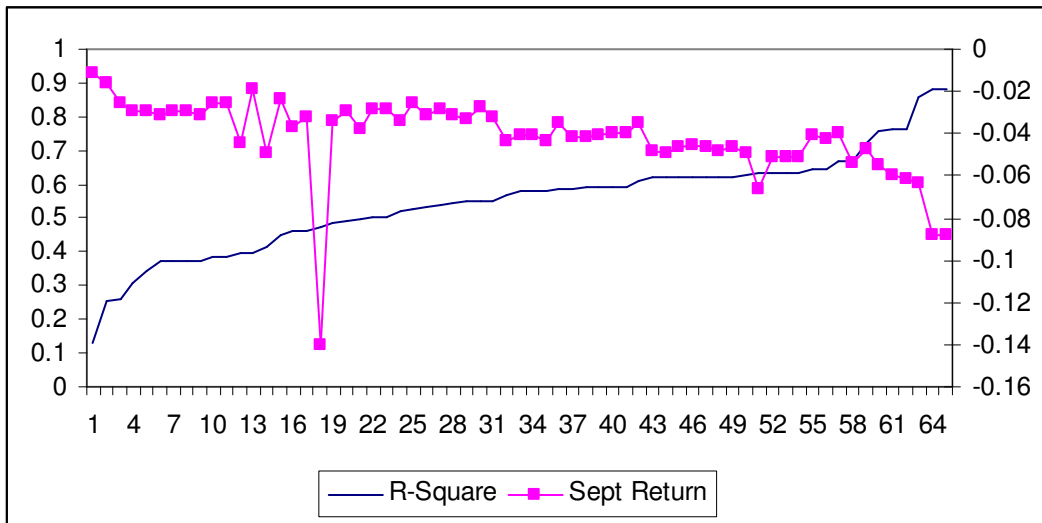


Exhibit 6: R-Square Frequencies

R-Square Range	Frequency	Average
0.130-0.399	13	-2.67%
0.400-0.599	28	-3.89%
Over 0.600	24	-5.32%

Exhibit 7: Regression of Amaranth Returns Against Major Hedge Fund Indices

Regression: Amaranth Return = α + β*Index Return + ϵ				
Index = CISDM Equal Weighted Hedge Fund Index				
Regression Statistics				
R Square	0.07			
Adjusted R Square	0.05			
	Coefficients	Standard Error	t Stat	P-value
Intercept	-0.02	0.02	-0.99	0.33
Index Return	2.31	1.26	1.83	0.07
Index = CSFB Hedge Fund Index				
Regression Statistics				
R Square	0.08			
Adjusted R Square	0.06			
	Coefficients	Standard Error	t Stat	P-value
Intercept	-0.02	0.02	-1.12	0.27
Index Return	2.96	1.49	1.99	0.05
Index = HFR Fund Weighted Composite Index				
Regression Statistics				
R Square	0.06			
Adjusted R Square	0.04			
	Coefficients	Standard Error	t Stat	P-value
Intercept	-0.02	0.02	-0.84	0.41
Index Return	2.03	1.23	1.65	0.11

Exhibit 8

